SIDARTHE Epidemic Model

A Compartmental Model of Infectious Disease Transmission

Language: Python
Application Domain: Mathematical Epidemiology
Date: March 2026

1. Introduction

The SIDARTHE model is an extended compartmental epidemiological framework designed to capture heterogeneous disease progression and detection dynamics. Originally proposed by Giordano et al. (2020), it partitions the population into eight mutually exclusive states rather than the three or four used by classical SIR and SEIR models. This granularity enables the model to distinguish between infected individuals who are undetected by the healthcare system and those who are formally diagnosed, as well as between individuals receiving standard care and those requiring intensive clinical intervention.

The Python class documented herein implements this model using an object-oriented design with discrete-time forward Euler integration. It includes a non-negativity guard on the susceptible compartment to prevent numerical artefacts, and provides integrated visualisation via Matplotlib. This document provides a systematic technical analysis of each component: its epidemiological meaning, mathematical formulation, and implementation details.

2. Model Overview and Compartmental Structure

2.1 The Eight Compartments

The SIDARTHE acronym encodes eight population states. Each individual in the model occupies exactly one compartment at any given time step, and the sum of all compartment populations equals the total normalised population N:

$$S + I + D + A + R + T + H + E = N, \ \forall t$$

Symbol	Full Name	Epidemiological Meaning
S	Susceptible	Uninfected individuals who may contract the disease upon contact with infectious compartments
I	Infected (undetected)	Infected but not formally diagnosed; infectious and unaware of their status
D	Diagnosed	Infected and confirmed positive via testing; assumed to be isolated and less infectious
A	Ailing (undetected)	Symptomatic but undiagnosed; still circulating in the community
R	Recognised	Symptomatic and formally diagnosed; receives standard medical care
T	Threatened	Critically ill patients requiring intensive care (ICU); life-threatening condition
H	Healed	Recovered individuals who are no longer infectious and have acquired immunity
E	Extinct	Disease-related fatalities; an absorbing state representing cumulative mortality

2.2 Key Modelling Extensions over SEIR

The SIDARTHE model introduces three structural extensions beyond the classical SEIR formulation:

Detection asymmetry: Undetected compartments (I and A) transmit disease at different rates from detected compartments (D and R), reflecting the lower infectiousness of individuals under isolation or medical supervision.

Severity stratification: The Threatened (T) compartment captures ICU-level disease burden separately from the general Recognised (R) pool, enabling direct modelling of healthcare system capacity pressure.

Explicit mortality: The Extinct (E) compartment accumulates disease-attributed deaths from the T compartment only. This reflects the modelling assumption that mortality occurs exclusively at the most severe disease stage.

3. Mathematical Foundation

3.1 Transmission Force and Non-Negativity Guard

New infections at each time step are driven by the total *force of infection*, defined as the weighted sum of infectious contacts from all four transmitting compartments:

$$\Lambda(t) = \beta 1 \cdot I(t) + \beta 2 \cdot D(t) + \beta 3 \cdot A(t) + \beta 4 \cdot R(t)$$

The implementation includes a **non-negativity guard** that prevents S from becoming negative when the force of infection exceeds the remaining susceptible pool. When `S[i] < Λ(t)`, the susceptible compartment is clamped to zero and new infections are computed using the residual S value from that step, effectively halting further depletion of S.

```
if self.S[i] < self.b1*self.I[i] + self.b2*self.D[i] + self.b3*self.A[i] +
self.b4*self.R[i]:
    self.S.append(0)   # Clamp: susceptible pool exhausted
    self.I.append(self.I[i] + self.S[i] - ...)   # Use remaining S
else:
    self.S.append(self.S[i] - self.b1*self.I[i] - ...)   # Normal update
```

3.2 Discrete-Time Difference Equations

The full system of forward Euler difference equations, applied at each time step i, is given below. Both branches of the non-negativity guard share identical equations for all compartments except S and I:

```
# Susceptible: reduced by total force of infection
S[t+1] = S[t] - β1·I[t] - β2·D[t] - β3·A[t] - β4·R[t]

# Infected (undetected): gain from S infections; lose to diagnosis, ailing, recovery
I[t+1] = I[t] + (β1·I[t] + β2·D[t] + β3·A[t] + β4·R[t]) - (α1 + γ1 + ρ1)·I[t]

# Diagnosed: gain from I via testing (α1); lose to recognition and natural recovery
D[t+1] = D[t] + α1·I[t] - (γ2 + ρ2)·D[t]

# Ailing (undetected): gain from I symptom onset (γ1); lose to
recognition/ICU/recovery
A[t+1] = A[t] + γ1·I[t] - (ρ3 + μ1 + α2)·A[t]

# Recognised: gain from D (γ2) and A (α2); lose to ICU admission and recovery
R[t+1] = R[t] + γ2·D[t] + α2·A[t] - (ρ4 + μ2)·R[t]

# Threatened (ICU): gain from A (μ1) and R (μ2); lose to recovery and death
T[t+1] = T[t] + μ1·A[t] + μ2·R[t] - (ρ5 + τ)·T[t]

# Healed: cumulative recoveries from I, D, A, R, T
H[t+1] = H[t] + ρ1·I[t] + ρ2·D[t] + ρ3·A[t] + ρ4·R[t] + ρ5·T[t]

# Extinct: cumulative deaths from T only
E[t+1] = E[t] + τ·T[t]
```

4. Model Parameters

4.1 Transmission Parameters

Four transmission rate parameters govern the flow from the susceptible compartment into the infected pool. Each parameter represents the per-contact probability of infection, scaled by the contact frequency with that compartment.

Symbol	Code Name	Source Compartment	Interpretation	Reference Value
β1	`b1`	I (Infected)	Transmission rate from undetected infected individuals	0.1779
β2	`b2`	D (Diagnosed)	Transmission rate from diagnosed cases; lower than b1, reflecting isolation compliance	0.0048
β3	`b3`	A (Ailing)	Transmission rate from symptomatic undetected individuals	0.0048
β4	`b4`	R (Recognised)	Transmission rate from formally managed cases	0.3926

4.2 Disease Progression Parameters

These parameters control the rates at which individuals transition between clinical states within the infected population. The reciprocal of each rate gives the mean sojourn duration in the source compartment, expressed in whatever time unit the simulation's time vector represents.

Symbol	Code Name	Transition	Interpretation	Ref. Value	Mean Duration (time steps)
α1	`a1`	I → D	Rate of diagnosis of undetected infected individuals	0.1238	1/0.1238 ≈ 8.08
α2	`a2`	A → R	Rate of clinical recognition of ailing (symptomatic) cases	0.1799	1/0.1799 ≈ 5.56
γ1	`g1`	I → A	Rate of symptom onset in undetected infected	0.1480	1/0.1480 ≈ 6.76
γ2	`g2`	D → R	Rate of clinical progression of diagnosed to recognised	0.0430	1/0.0430 ≈ 23.26
μ1	`u1`	A → T	Rate of severe progression from ailing to threatened	0.0255	1/0.0255 ≈ 39.22
μ2	`u2`	R → T	Rate of severe progression from recognised to threatened	0.0336	1/0.0336 ≈ 29.76

4.3 Recovery and Mortality Parameters

Recovery rates determine the rate of flow into the Healed (H) compartment from each infected state. The single mortality parameter τ governs the death rate exclusively from the Threatened (T)

compartment.

Symbol	Code Name	Source	Interpretation	Ref. Value
ρ1	`r1`	I	Natural recovery rate of undetected infected	0.0499
ρ2	`r2`	D	Recovery rate of diagnosed cases under monitoring	0.0499
ρ3	`r3`	A	Recovery rate of ailing (symptomatic, undetected) patients	0.1480
ρ4	`r4`	R	Recovery rate of formally recognised patients	0.1480
ρ5	`r5`	T	Recovery rate of ICU (threatened) patients	0.1480
τ	`t`	T	Mortality rate from ICU; sole source of deaths in this model	0.0127

4.4 Complete Parameter Reference

The following table consolidates all parameters as passed to the reference instantiation in the __main__ block:

```
s = SIDARTHE(T, N=1, I=0.0005, D=0, A=0, R=0, T1=0, H=0, E=0,
             r1=0.0499, r2=0.0499, r3=0.1480, r4=0.1480, r5=0.1480,
             b1=0.1779, b2=0.0048, b3=0.0048, b4=0.3926,
             a1=0.1238, a2=0.1799, g1=0.1480, g2=0.0430,
             u1=0.0255, u2=0.0336, t=0.0127)
```

5. Class Architecture

5.1 Class Overview

The model is encapsulated in a single Python class named `SIDARTHE`. Following the same object-oriented pattern as the companion SEIR implementation, the class exposes three public methods: a constructor for state initialisation, a numerical integration method, and a visualisation method.

```
class SIDARTHE:
    def __init__(self, T, N, I, D, A, R, T1, H, E, r1..r5, b1..b4, a1, a2, g1, g2, u1,
u2, t)
    def calc(self)      # Numerical integration with non-negativity guard
    def plot(self)      # Eight-curve visualisation via Matplotlib
```

5.2 Constructor: __init__

The constructor receives 26 arguments — all epidemiological parameters plus the time vector — and stores them as instance attributes. Each of the eight compartment histories is initialised as a single-

element list containing the seed value at t = 0.

```
def __init__(self, T, N, I, D, A, R, T1, H, E,
             r1, r2, r3, r4, r5,   # Recovery rates (ρ)
             b1, b2, b3, b4,        # Transmission rates (β)
             a1, a2,                # Diagnosis/recognition rates (α)
             g1, g2,                # Symptom onset / progression rates (γ)
             u1, u2,                # ICU admission rates (μ)
             t):                    # Mortality rate (τ)
    self.N  = N
    self.I  = [I]; self.D = [D]; self.A = [A]; self.R = [R]
    self.T1 = [T1]; self.H = [H]; self.E = [E]
    self.S  = [N - I - D - A - R - T1 - H - E]  # Conservation identity
    # ... store all rate parameters ...
    self.T  = T
```

Compared to the SEIR implementation, the initial susceptible calculation here is fully consistent: $S_0 = N - I_0 - D_0 - A_0 - R_0 - T_0 - H_0 - E_0$. With the reference values, all non-S compartments except I are zero, so $S_0 = 1 - 0.0005 = 0.9995$.

5.3 Computational Method: calc()

The `calc()` method drives the simulation forward by iterating over the time vector and applying the SIDARTHE difference equations. The method is structured around a conditional non-negativity guard evaluated at every time step.

5.3.1 Idempotency Guard

As in the SEIR implementation, the method opens with a check `if len(self.T) == len(self.S)` that short-circuits execution if the simulation has already been completed, preventing list over-extension on repeated calls.

5.3.2 Non-Negativity Guard

At each time step, the method checks whether the total force of infection would exceed the current susceptible pool. This prevents S from becoming negative, which would be epidemiologically nonsensical and numerically destabilising:

```
# Branch 1: Susceptible pool would be exhausted
if S[i] < b1·I[i] + b2·D[i] + b3·A[i] + b4·R[i]:
    S[t+1] = 0
    I[t+1] = I[i] + S[i] - (a1 + g1 + r1)·I[i]   # Uses residual S[i]

# Branch 2: Normal update
else:
    S[t+1] = S[i] - b1·I[i] - b2·D[i] - b3·A[i] - b4·R[i]
    I[t+1] = I[i] + (b1·I[i] + b2·D[i] + b3·A[i] + b4·R[i]) - (a1+g1+r1)·I[i]
```

Equations for D, A, R, T, H, and E are structurally identical in both branches, as the pool exhaustion event only directly affects S and the new-infection term in I.

5.4 Visualisation Method: plot()

The `plot()` method renders eight time-series curves simultaneously, one per compartment. It follows the same lazy-evaluation pattern as the SEIR implementation: if `calc()` has not yet been called (detected by `len(self.S) == 1`), it is invoked automatically before plotting.

```
def plot(self):
    import matplotlib.pyplot as plt
    import matplotlib as mpl
    mpl.rcParams['axes.unicode_minus'] = False

    if len(self.S) == 1:
        self.calc()    # Lazy evaluation

    plt.figure()
    plt.plot(self.T, self.S,  color='b',           label='Susceptible')
    plt.plot(self.T, self.I,  color='r',           label='Infected')
    plt.plot(self.T, self.D,  color='y',           label='Diagnosed')
    plt.plot(self.T, self.A,  color='c',           label='Ailing')
    plt.plot(self.T, self.R,  color='m',           label='Recognized')
    plt.plot(self.T, self.T1, color='g',           label='Threatened')
    plt.plot(self.T, self.H,  color='k',           label='Healed')
    plt.plot(self.T, self.E,  color='xkcd:orange', label='Extinct')
    plt.legend(); plt.xlabel('Time')
    plt.ylabel('Proportion of population'); plt.show()
```

Note: The internal variable name `T1` is used to store the Threatened compartment history to avoid a naming conflict with Python's built-in type annotation convention and the parameter `T` used for the time vector. In the plot legend and all scientific literature, this compartment is denoted simply as T.

6. Execution and Reference Run

6.1 Entry Point

The `if __name__ == '__main__'` guard at the module level ensures that the reference simulation runs only when the script is executed directly. This preserves the class's usability as an importable module for parameter sweeps, calibration pipelines, or ensemble studies.

```
if __name__ == '__main__':
    T = [i for i in range(150)]         # 150 discrete time steps
    s = SIDARTHE(T, 1, 0.0005, 0, 0, 0, 0, 0, 0,
                 0.0499, 0.0499, 0.1480, 0.1480, 0.1480,  # r1-r5
                 0.1779, 0.0048, 0.0048, 0.3926,           # b1-b4
                 0.1238, 0.1799, 0.1480, 0.0430,           # a1,a2,g1,g2
                 0.0255, 0.0336, 0.0127)                   # u1,u2,t
    s.plot()
```

6.2 Colour Legend and Expected Output

The simulation produces a multi-curve time-series plot. The following table documents the colour encoding and expected qualitative trajectory of each compartment:

Colour	Compartment	Description
Blue	**Susceptible (S)**	Monotonically decreasing sigmoid; initial value ~0.9995, declining as the epidemic spreads.
Red	**Infected (I)**	Initial unimodal peak representing the undetected infectious wave; drives most early transmission.
Yellow	**Diagnosed (D)**	Lagged peak following I; represents confirmed cases identified through testing.
Cyan	**Ailing (A)**	Symptomatic undetected; peaks between I and R curves.
Magenta	**Recognised (R)**	Formally managed symptomatic cases; peaks after A.
Green	**Threatened (T)**	ICU patients; small magnitude but critical for mortality and healthcare capacity estimation.
Black	**Healed (H)**	Monotonically increasing; largest final value, reflecting the dominant outcome.
Orange	**Extinct (E)**	Monotonically increasing from zero; small absolute value reflecting ICU mortality rate τ.

7. Comparison with the Companion SEIR Model

Feature	SEIR	SIDARTHE
Compartments	4 (S, E, I, R)	8 (S, I, D, A, R, T, H, E)
Parameters	10 (including T vector)	26 (including T vector)
Transmission sources	I and E	I, D, A, R
Detection modelling	Not explicit	Explicit: undetected vs. diagnosed streams
ICU dynamics	Not represented	Dedicated Threatened (T) compartment
Mortality	Not modelled	Explicit Extinct (E) compartment; ICU-only
Non-negativity guard	Not present	Applied to S at every time step
Computational complexity	O(n) iterations, 4 updates/step	O(n) iterations, 8 updates/step + branch
Primary application	General epidemic modelling, pedagogy	Epidemics with heterogeneous detection and severity stratification

8. Summary

The SIDARTHE class provides a high-fidelity, object-oriented implementation of an eight-compartment epidemic model. By distinguishing between detected and undetected infected populations, stratifying disease severity into standard and critical care categories, and explicitly tracking both recoveries and fatalities, the model captures a substantially richer epidemiological picture than classical SEIR formulations.

The implementation's three-method interface maintains clean separation between state initialisation, numerical computation, and result visualisation. The non-negativity guard on the susceptible compartment adds numerical robustness without materially altering the model's dynamics under typical parameterisations. The lazy evaluation pattern in plot() ensures that the computation is triggered automatically when needed, reducing the cognitive load on the caller.

For research applications, this implementation serves as a transparent and inspectable baseline. Recommended extensions include: time-varying transmission parameters calibrated to intervention timelines, stochastic formulations to propagate parameter uncertainty, age-stratified contact matrices, and replacement of the forward Euler scheme with adaptive ODE solvers for improved numerical fidelity.

References

[1] Giordano, G., Blanchini, F., Bruno, R., Colaneri, P., Di Filippo, A., Di Matteo, A., & Colaneri, M. (2020). Modelling the COVID-19 epidemic and implementation of population-wide interventions in Italy. *Nature Medicine*, 26(6), 855–860.

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[4] Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3), 90–95.